38 Two neutral atoms are isotopes of the same element.

Which statement about the atoms is correct?

A They have a different number of neutrons and a different number of electrons.

B They have a different number of neutrons and the same number of protons.

C They have the same number of neutrons and a different number of electrons.

D They have the same number of neutrons and the same number of protons.

39 What is the composition of a meson?

A 1 quark and 1 antiquark

B 1 quark and 2 antiquarks

C 2 quarks and 1 antiquark

D 3 quarks

40 A nucleus of carbon-11 contains 6 protons and 5 neutrons.

The nucleus of carbon-11 decays by β^+ emission.

What is the total number of up and down quarks in the product of the decay of this nucleus?

	up quarks	down quarks
A	15	18
B	16	17
C	17	16
D	18	15